

A proof of Conjecture 00000006420

Two unions of two unit disks with the same isoperimetric deficit and different Fraenkel asymmetry

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1 The conjecture

Conjecture 00000006420 reads (English version; the Chinese version agrees):

Definition: The isoperimetric deficit and asymmetry are two layers. *Conjecture:* There exist two sets with the same deficit but different asymmetry, and the separation is realized by an explicit two-circle union with the same deficit but different asymmetry.

This is an existence claim. We exhibit two explicit sets in the plane, each the union of two disjoint closed unit disks, prove that they have the same isoperimetric deficit (namely $\sqrt{2} - 1$), and prove that their Fraenkel asymmetries differ. The main results are formalized in Lean 4 / Mathlib for the perimeter $P(E) = \mathcal{H}^1(\partial E)$ defined below. The De Giorgi (distributional) perimeter is not defined in Lean: `perimeter_axiomatic` is a conditional statement about any translation-invariant perimeter that is additive on disjoint compact sets, and its application to the De Giorgi perimeter (and that perimeter's agreement with $\mathcal{H}^1(\partial E)$ for these sets) is a standard fact that is not formalized; see the conventions paragraph at the end.

2 Reading and definitions

We work in the Euclidean plane $\mathbb{R}^2$ (Mathlib: `EuclideanSpace R (Fin 2)`) with Lebesgue measure $|\cdot|$. “Asymmetry” is read as the Fraenkel asymmetry, the asymmetry index that is paired with the isoperimetric deficit in the quantitative isoperimetric inequality $\alpha(E)^2 \leq C \delta(E)$. We use the normalizations of Cicalese and Leonardi (*A selection principle for the sharp quantitative isoperimetric inequality*, <https://arxiv.org/abs/1007.3899>, Section 2), which follow Fusco, Maggi and Pratelli. A “two-circle union” is read as the union of two closed disks. Two circles as curves have area zero, so their deficit is not defined.

Definition 1. For $E \subset \mathbb{R}^2$ let $r_E = \sqrt{|E|/\pi}$, the radius of the disk B_E with $|B_E| = |E|$.

- *Perimeter:* $P(E) = \mathcal{H}^1(\partial E)$, the one-dimensional Hausdorff measure (length) of the topological boundary. In Lean this is `perimeter E := muH[1] (frontier E)`. Mathlib's $\mu_H[1]$ needs no normalizing constant in dimension one; on $\mathbb{R}$ it equals Lebesgue measure (`hausdorffMeasure_real`).
- *Isoperimetric deficit:* $\delta(E) = \frac{P(E) - P(B_E)}{P(B_E)} = \frac{P(E)}{P(B_E)} - 1$, with $B_E = \bar{B}(0, r_E)$ (`deficit`).
- *Fraenkel asymmetry:* $\alpha(E) = \inf_{x \in \mathbb{R}^2} \frac{|E \triangle \bar{B}(x, r_E)|}{|E|}$ (`asymmetry`).

3 The two sets

Let $D_a = \bar{B}((a, 0), 1)$ be the closed unit disk centred at $(a, 0)$, and put

$$E_{\text{near}} = D_0 \cup D_{5/2}, \quad E_{\text{far}} = D_0 \cup D_{10}.$$

In both unions the centres are more than 2 apart, so the two disks are disjoint (`disj`). Hence $|E_{\text{near}}| = |E_{\text{far}}| = 2\pi$, $r_E = \sqrt{2}$ in both cases, and every comparison disk $\bar{B}(x, \sqrt{2})$ has area 2π (`area_near`, `area_far`, `eqRadius_near`, `eqRadius_far`, `vol_comparison`).

4 Same deficit

Lemma 2 (`frontier_union_of_closed`). *If A, B are closed and disjoint, then $\partial(A \cup B) = \partial A \cup \partial B$.*

Proof. The inclusion $\subseteq$ holds in general. Conversely, let $p \in \partial A$. Then $p \in A$, so $p \in \overline{A \cup B}$. If p were interior to $A \cup B$, then $U = \text{int}(A \cup B) \setminus B$ would be an open neighbourhood of p (because $p \notin B$ and B is closed) contained in A , contradicting $p \in \partial A$. $\square$

Let $L = \mathcal{H}^1(S^1)$ be the length of the unit circle.

Lemma 3 (`perimeter_two`, `L_pos`, `L_lt_top`). *For $|a| > 2$ we have $P(D_0 \cup D_a) = 2L$, and $0 < L < \infty$. The proofs establish $L \leq 2\pi$ and $2 \leq L$ as intermediate steps.*

Proof. By the previous lemma and $\partial \bar{B}(c, 1) = S(c, 1)$, the boundary is the disjoint union of two unit circles. $\mathcal{H}^1$ is a Borel measure and is invariant under translations (isometries), so $P = 2L$. For the upper bound, every point of the unit circle in $\mathbb{C} \cong \mathbb{R}^2$ is $e^{i\theta}$ with $\theta = \arg z \in [-\pi, \pi]$, and $\theta \mapsto e^{i\theta}$ is 1-Lipschitz, because $|e^{is} - e^{it}| = |e^{i(s-t)} - 1| \leq |s - t|$. Lipschitz maps do not increase $\mathcal{H}^1$, so $L \leq \mathcal{H}^1([-\pi, \pi]) = 2\pi$. For the lower bound, the projection $z \mapsto \text{Re } z$ is 1-Lipschitz and maps the circle onto $[-1, 1]$, so $2 = \mathcal{H}^1([-1, 1]) \leq L$. The identification $\mathbb{C} \cong \mathbb{R}^2$ is the isometry given by the orthonormal basis $(1, i)$ (`L_eq_complex`). $\square$

Theorem 4 (`deficit_two`). $\delta(E_{\text{near}}) = \delta(E_{\text{far}}) = \sqrt{2} - 1$.

Proof. We have $\partial \bar{B}(0, \sqrt{2}) = S(0, \sqrt{2}) = \sqrt{2} \cdot S(0, 1)$, and $\mathcal{H}^1$ scales by the factor $\sqrt{2}$ under this dilation (`Measure.hausdorffMeasure_smul0`). So $P(B_E) = \sqrt{2}L$ (`perimeter_comparison`), and $\delta = 2L/(\sqrt{2}L) - 1 = \sqrt{2} - 1$, using $0 < L < \infty$. $\square$

5 Different asymmetry

Theorem 5 (`asymmetry_far`). $\alpha(E_{\text{far}}) \geq 1$.

Proof. Fix x and let $B = \bar{B}(x, \sqrt{2})$. The disk B cannot meet both D_0 and D_{10} : otherwise $10 = |(0, 0) - (10, 0)| \leq 1 + 2\sqrt{2} + 1 < 5$ (`far_cases`). Say $B \cap D_{10} = \emptyset$; the other case is symmetric. Then $D_{10} \subseteq E \setminus B$ and $B \setminus D_0 \subseteq B \setminus E$, and these two sets are disjoint. So $|E \triangle B| \geq |D_{10}| + |B| - |D_0| = \pi + 2\pi - \pi = 2\pi$ (`symmDiff_lower`). Dividing by $|E| = 2\pi$ gives a quotient of at least 1 for every x , so the infimum is at least 1. $\square$

Theorem 6 (`near_value`, `asymmetry_near_le`, `asymmetry_near`). $\alpha(E_{\text{near}}) \leq 1 - \frac{1}{100} < 1$.

Proof. Take $x_0 = (2/5, 0)$, $B_0 = \bar{B}(x_0, \sqrt{2})$ and $S = \bar{B}((8/5, 0), 1/10)$. Then: $D_0 \subseteq B_0$ since $1 + \frac{2}{5} = \frac{7}{5} \leq \sqrt{2}$, because $\frac{49}{25} \leq 2$; $S \subseteq D_{5/2}$ since $\frac{1}{10} + \frac{9}{10} = 1$; and $S \subseteq B_0$ since $\frac{1}{10} + \frac{6}{5} = \frac{13}{10} \leq \sqrt{2}$. So $K = D_0 \cup S \subseteq E_{\text{near}} \cap B_0$ is a disjoint union with $|K| = \pi + \pi/100$. Since $E \triangle B_0 \subseteq (E \setminus K) \cup (B_0 \setminus K)$, we get

$$|E_{\text{near}} \triangle B_0| \leq (2\pi - |K|) + (2\pi - |K|) = 2\pi - \frac{\pi}{50}.$$

The quotient at x_0 is therefore at most $1 - \frac{1}{100}$. The quotients are non-negative, so the infimum is at most this value. $\square$

Theorem 7 (**separation, conjecture6420**). E_{near} and E_{far} are unions of two disjoint closed unit disks. They have equal area 2π , equal perimeter $2L \in (0, \infty)$, and equal deficit $\sqrt{2} - 1$, while $\alpha(E_{\text{near}}) < 1 \leq \alpha(E_{\text{far}})$. Hence the conjecture holds.

Other conventions (normalisations, perimeter_axiomatic). (i) Any deficit that is a function of $(P(E), |E|)$, for example $P(E)^2/(4\pi|E|) - 1$ or $P(E) - 2\sqrt{\pi|E|}$, agrees on the two sets. (ii) Any positive rescaling of α still separates them. This covers the normalization $|E \triangle (x + B_E)|/r_E^2$, which is π times ours. (iii) The De Giorgi (distributional) perimeter is the convention of the cited paper. Lean proves that *every* perimeter functional that is translation invariant and additive on disjoint compact sets, a pair of properties that the De Giorgi perimeter has, takes the same value on E_{near} and E_{far} . So every deficit built from such a perimeter and the area agrees. That the De Giorgi perimeter has these two properties, and that it equals $\mathcal{H}^1(\partial E)$ for these smooth sets, are standard facts that are not formalized here.

6 Lean formalization and axiom audit

The file `lean/Conjecture6420/Basic.lean` (Lean 4.33.1, Mathlib v4.33.1, single import `Mathlib`) contains all definitions and proofs, in namespace `C6420`. Main theorems: `separation`, `conjecture6420`, `normalisations` and `perimeter_axiomatic`. For each of them, `#print axioms` reports exactly `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom. The value of L ($= 2\pi$) is never needed: only $0 < L < \infty$ is used.